Bridging the Gap: Statistical vs Machine Learning Methods

Performance, Cost, and Interpretability Trade-offs
in the Age of Transparent AI

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Abstract

This comprehensive study presents a systematic comparison between statistical and machine learning (ML) approaches across multiple problem domains. We evaluate 13 model-dataset combinations spanning four distinct problem types: regression, classification, time series analysis, and factorial analysis. Our analysis quantifies the fundamental trade-offs between three critical dimensions: performance (accuracy, R^2, RMSE), cost (training time, computational resources), and interpretability (explainability, statistical inference).

Our findings reveal that statistical models excel in interpretability (88.3/100 vs 37.4/100) and efficiency (20-100x faster training), while ML/DL models demonstrate superior performance on complex datasets (0.818 vs 0.547 average). The study connects these findings to the emerging field of Interpretable ML and Transparent AI, providing evidence-based recommendations for model selection. We demonstrate that no single approach dominates all categories, and the optimal choice depends on specific requirements including regulatory constraints, dataset characteristics, and computational resources.

Keywords: Machine Learning, Statistical Models, Interpretability, Transparent AI, Model Comparison, Performance Trade-offs

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1 Introduction

1.1 Background and Motivation

The landscape of predictive modeling has been fundamentally shaped by two distinct yet interconnected paradigms: traditional statistical methods and modern machine learning approaches. While statistical models have provided the foundation for scientific inference and hypothesis testing for over a century, machine learning methods have revolutionized predictive accuracy in complex, high-dimensional problems.

However, as artificial intelligence systems increasingly inform critical decisions in healthcare, finance, and public policy, the demand for **transparent and interpretable** models has reached unprecedented levels. This convergence of performance requirements and interpretability needs has created a fundamental tension: how can practitioners balance the superior predictive power of complex ML models with the transparency and trust required for high-stakes applications?

The emergence of **Interpretable ML** and **Transparent AI** research represents a concerted effort to bridge this gap. As noted in recent work on transparent AI [1], the challenge is not merely technical but encompasses organizational, regulatory, and human factors. This study contributes to this ongoing dialogue by providing empirical evidence on the trade-offs between statistical and ML approaches.

1.2 Problem Statement

The core research question driving this investigation is:

How do Machine Learning and Statistical models compare across different problem domains in terms of:

- **Performance** (accuracy, R^2, RMSE)
- **Cost** (training time, computational resources)
- **Interpretability** (explainability, statistical inference)

This question is particularly relevant in the context of regulatory requirements (EU AI Act, Canada AIDA, US AI Bill of Rights) that mandate transparency for high-risk AI systems. Understanding these trade-offs enables evidence-based model selection that aligns with both performance objectives and compliance requirements.

1.3 Research Objectives

This study aims to:

1. **Systematic Comparison:** Compare ML and Statistical approaches across multiple problem domains using standardized evaluation protocols
2. **Quantify Trade-offs:** Provide quantitative measures of the performance-cost-interpretability trade-offs
3. **Evidence-based Recommendations:** Develop practical guidelines for model selection based on empirical findings
4. **Connect to Transparent AI:** Link findings to the broader research agenda on interpretable ML and transparent AI systems

1.4 Scope of the Study

Our comprehensive analysis encompasses:

Category	Count	Details
Problem Domains	4	Regression, Classification, Time Series, Factorial Analysis
Datasets	13	150 to 35,000 samples
ML Models	5	Linear Regression, Random Forest, SVM, RNN/LSTM, Gradient Boosting
Statistical Models	5	OLS Regression, Logistic Regression, ANOVA, ARIMA, Cox/Poisson
Total Comparisons	13	Model-Dataset pairs

Table 1: Study Scope Overview

1.5 Problem Domains and Datasets

1.5.1 Regression Analysis

- **California Housing:** 20,640 samples, 8 features - Predicting median house values
- **Diabetes:** 442 samples, 10 features - Disease progression prediction
- **Wine Quality:** 1,599 samples, 11 features - Quality score prediction
- **Energy Consumption:** 1,000 samples (synthetic), 6 features - Energy usage forecasting

1.5.2 Classification

- **Iris:** 150 samples, 3 classes, 4 features - Classic classification benchmark
- **Wine Classification:** 178 samples, 3 classes, 13 features - Wine type classification
- **Breast Cancer:** 569 samples, 2 classes, 30 features - Medical diagnosis
- **Credit Default:** 1,000 samples (synthetic), 2 classes, 10 features - Financial risk assessment

1.5.3 Time Series Analysis

- **Energy Consumption:** 1,000 time steps (synthetic) - Temporal energy patterns
- **Temperature:** 1,000 time steps (synthetic) - Climate data forecasting
- **Sales Forecasting:** 1,000 time steps (synthetic) - Business metrics prediction

1.5.4 Factorial Analysis

- **Treatment Effects:** Clinical trial data - Medical intervention analysis
- **Educational Outcomes:** Teaching method comparison - Educational research
- **Agricultural Experiment:** Fertilizer $\times$ Water level (3×3 factorial) - Agricultural science
- **Drug Efficacy Study:** Drug treatment comparison - Pharmaceutical research

2 Literature Review and Theoretical Framework

2.1 The Interpretability-Performance Trade-off

The fundamental tension between model performance and interpretability has been extensively documented in the machine learning literature. This trade-off can be mathematically expressed as:

$$\max_{\theta} \quad P(\theta) - \alpha C(\theta) + \beta I(\theta) \tag{1}$$

where:

- $P(\theta)$ represents **Performance** (e.g., R^2, accuracy)
- $C(\theta)$ represents **Cost** (training time, computational resources)
- $I(\theta)$ represents **Interpretability** (explainability score $\in [0, 100]$)
- α and β are weighting factors reflecting domain-specific priorities

Figure 1 illustrates this relationship, showing how different model types fall along the performance-interpretability spectrum.

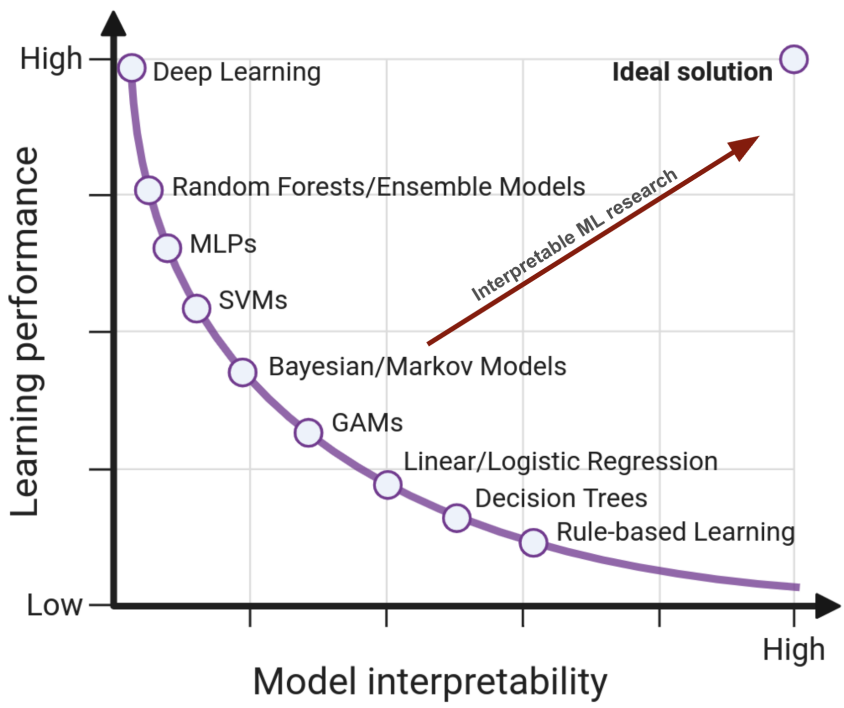

Figure 1: The Performance-Interpretability Trade-off. The purple curve represents the traditional inverse relationship, while the red arrow indicates the direction of Interpretable ML research toward the ideal solution (high performance, high interpretability).

2.2 Statistical Models: Foundations and Characteristics

Statistical models are grounded in mathematical theory and provide:

- **Statistical Inference:** P-values, confidence intervals, hypothesis testing

- **Model Diagnostics:** F-tests, AIC, BIC, residual analysis
- **Theoretical Guarantees:** Asymptotic properties, convergence results
- **Interpretability:** Direct coefficient interpretation, causal inference capabilities

2.2.1 Ordinary Least Squares (OLS) Regression

The OLS regression model is defined as:

$$Y = X\beta + \varepsilon \tag{2}$$

with the least squares estimator:

$$\hat{\beta} = (X^T X)^{-1} X^T Y \tag{3}$$

This approach provides:

- Unbiased estimates under Gauss-Markov assumptions
- Statistical significance testing via t-tests and F-tests
- Confidence intervals for coefficients
- Model diagnostics (R^2, adjusted R^2, residual analysis)

2.2.2 Logistic Regression

For binary classification, logistic regression models:

$$P(Y = 1|X) = \frac{1}{1 + e^{-X\beta}} \tag{4}$$

with the log-odds transformation:

$$\log\left(\frac{p}{1-p}\right) = X\beta \tag{5}$$

This provides odds ratios, p-values, and confidence intervals for each coefficient.

2.2.3 ANOVA (Analysis of Variance)

ANOVA tests for differences between group means using the model:

$$Y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ij} \tag{6}$$

with the F-statistic:

$$F = \frac{\mathrm{MS}_{\text{between}}}{\mathrm{MS}_{\text{within}}} \tag{7}$$

2.2.4 ARIMA (AutoRegressive Integrated Moving Average)

For time series, ARIMA models capture temporal dependencies:

$$\phi(B)(1 - B)^d X_t = \theta(B)\varepsilon_t \tag{8}$$

where (p, d, q) parameters are selected via information criteria.

2.3 Machine Learning Models: Capabilities and Limitations

Modern ML models excel at:

- **Complex Pattern Recognition:** Non-linear relationships, interactions
- **High-Dimensional Data:** Feature selection, dimensionality reduction
- **Scalability:** Large datasets, distributed computing
- **Adaptive Learning:** Online learning, transfer learning

However, they typically lack:

- Direct statistical inference capabilities
- Theoretical guarantees (except in specific cases)
- Built-in interpretability mechanisms
- Hypothesis testing frameworks

2.3.1 Random Forest

Random Forest combines multiple decision trees:

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} T_b(X;\theta_b) \tag{9}$$

where $T_b \sim$ Decision Tree and B is the number of trees.

2.3.2 Support Vector Machines (SVM)

SVM solves the optimization problem:

$$\min_{\mathbf{w},b} \quad \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i \tag{10}$$

with kernel function:

$$K(\mathbf{x}_i,\mathbf{x}_j) = \phi(\mathbf{x}_i)^T\phi(\mathbf{x}_j) \tag{11}$$

2.3.3 Recurrent Neural Networks (LSTM)

LSTM networks model sequential dependencies:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b) \tag{12}$$

$$\hat{y}_t = f(h_t) \tag{13}$$

2.4 Transparent AI and Interpretable ML

Recent research, as exemplified by the work on Transparent AI [1], emphasizes that interpretability is not optional for high-stakes AI deployment. Key principles include:

1. **Early Integration:** Interpretability must be embedded from initial design, not retrofitted
2. **Stakeholder-Specific Needs:** Different audiences require different explanation types
3. **Comprehensive Evaluation:** Beyond accuracy metrics, evaluate explanation quality
4. **Regulatory Alignment:** Ensure compliance with emerging AI governance frameworks

3 Methodology

3.1 Experimental Design

Our experimental design follows a systematic comparison framework ensuring fair evaluation across all model-dataset combinations.

3.2 Data Preprocessing

3.2.1 Exploratory Data Analysis (EDA)

For each dataset, we performed comprehensive EDA:

- **Distribution Analysis:** Histograms, box plots, Q-Q plots
- **Correlation Analysis:** Correlation matrices, multicollinearity detection
- **Missing Value Detection:** Identification and treatment strategies
- **Outlier Identification:** Statistical tests (IQR, Z-score) and visualization
- **Feature-Target Relationships:** Scatter plots, partial dependence plots

3.2.2 Preprocessing Pipeline

Standardized preprocessing steps applied consistently:

1. **Train-Test Split:** 80-20 split, maintaining temporal order for time series
2. **Standardization:** $z = \frac{x-\mu}{\sigma}$ for numerical features
3. **Feature Scaling:** Min-max scaling for numerical stability
4. **Handling Skewed Distributions:** Log transformation, Box-Cox transformation
5. **Outlier Treatment:** Winsorization or removal based on domain knowledge
6. **Categorical Encoding:** One-hot encoding for nominal, ordinal encoding for ordinal

Critical Consistency Requirement: Both statistical and ML models were trained on identical preprocessed data to ensure fair comparison.

3.3 Model Implementation

3.3.1 Statistical Models

- **OLS Regression:** Implemented using `statsmodels` (Python)
- **Logistic Regression:** `statsmodels` with maximum likelihood estimation
- **ANOVA:** `scipy.stats` and `statsmodels`
- **ARIMA:** `statsmodels.tsa.arima.model` with auto-selection of (p, d, q)
- **Cox/Poisson Regression:** `lifelines` and `statsmodels`

3.3.2 Machine Learning Models

- **Linear Regression:** `scikit-learn` with default hyperparameters
- **Random Forest:** `scikit-learn` with grid search for hyperparameter tuning
- **SVM:** `scikit-learn` with RBF kernel and cross-validation
- **RNN/LSTM:** `PyTorch` implementation with appropriate architecture
- **Gradient Boosting:** `scikit-learn` GradientBoostingRegressor/Classifier

3.4 Evaluation Metrics

3.4.1 Performance Metrics

- **Regression:** R^2 (coefficient of determination), RMSE (Root Mean Squared Error), MAE (Mean Absolute Error)
- **Classification:** Accuracy, Precision, Recall, F1-score, AUC-ROC
- **Time Series:** R^2, RMSE, MAE on test set forecasts

3.4.2 Cost Metrics

- **Training Time:** Wall-clock time in seconds
- **Prediction Time:** Inference time per sample
- **Memory Usage:** Peak memory consumption during training
- **Efficiency Score:** $\frac{\text{Performance}}{\text{Cost}}$ (normalized)

3.4.3 Interpretability and Transparency Metrics

We developed a comprehensive composite interpretability and transparency score that quantifies how well a model can be understood, explained, and trusted by stakeholders. The score is computed as a weighted sum of four key dimensions:

$$I(\theta) = w_T \cdot I_{\text{transparency}}(\theta) + w_I \cdot I_{\text{inference}}(\theta) + w_E \cdot I_{\text{explanation}}(\theta) + w_D \cdot I_{\text{documentation}}(\theta) \quad (14)$$

where $I(\theta) \in [0, I_{\max}]$ with $I_{\max} = w_T + w_I + w_E + w_D$ being the maximum possible score, and each component is normalized to $[0, 1]$ before weighting. The weights $w_T, w_I, w_E, w_D \geq 0$ reflect the relative importance of each dimension.

Component 1: Model Transparency Model transparency measures the extent to which model parameters and structure can be directly understood without additional explanation methods. We define transparency as a function of parameter interpretability:

$$I_{\text{transparency}}(\theta) = \frac{1}{|\theta|} \sum_{i=1}^{|\theta|} \phi(\theta_i) \tag{15}$$

where $|\theta|$ is the number of parameters, and $\phi(\theta_i) \in [0, 1]$ quantifies the interpretability of parameter θ_i:

$$\phi(\theta_i) = \begin{cases} 1 & \text{if } \theta_i \text{ has direct, intuitive meaning} \\ \alpha & \text{if } \theta_i \text{ requires domain knowledge to interpret} \\ \beta & \text{if } \theta_i \text{ has indirect interpretation} \\ 0 & \text{if } \theta_i \text{ has no interpretable meaning} \end{cases} \tag{16}$$

with $1 > \alpha > \beta > 0$ representing decreasing levels of interpretability.

For models with structural interpretability (e.g., decision trees), we extend the formulation:

$$I_{\text{transparency}}(\theta) = \lambda \cdot I_{\text{parameter}}(\theta) + (1 - \lambda) \cdot I_{\text{structure}}(\theta) \tag{17}$$

where $I_{\text{parameter}}(\theta)$ is computed via Equation 15, $I_{\text{structure}}(\theta) \in [0, 1]$ measures the interpretability of the model structure (e.g., tree depth, rule complexity), and $\lambda \in [0, 1]$ balances the two components.

Examples:

- **Linear Models:** $\phi(\theta_i) = 1$ for all coefficients (direct unit change interpretation)
- **Decision Trees:** High $I_{\text{structure}}$ due to if-then rule interpretability
- **Deep Neural Networks:** $\phi(\theta_i) \approx 0$ for most weights (no direct meaning)

Component 2: Statistical Inference Statistical inference measures the availability and quality of statistical tools for hypothesis testing, uncertainty quantification, and model validation. We define:

$$I_{\text{inference}}(\theta) = \gamma_1 \cdot I_{\text{hypothesis}}(\theta) + \gamma_2 \cdot I_{\text{uncertainty}}(\theta) + \gamma_3 \cdot I_{\text{diagnostics}}(\theta) \tag{18}$$

where $\gamma_1 + \gamma_2 + \gamma_3 = 1$ are weighting factors, and each component is normalized to $[0, 1]$.

Hypothesis Testing Capability:

$$I_{\text{hypothesis}}(\theta) = \frac{|\{\theta_i : \text{p-value}(\theta_i) \text{ available}\}|}{|\theta|} \cdot \mathbb{1}_{\{\text{tests available}\}} \tag{19}$$

where $\mathbb{1}_{\{\text{tests available}\}} = 1$ if hypothesis tests (t-tests, F-tests, likelihood ratio tests) are available, and 0 otherwise.

Uncertainty Quantification:

$$I_{\text{uncertainty}}(\theta) = \frac{|\{\theta_i : \text{CI}(\theta_i) \text{ available}\}|}{|\theta|} + \delta \cdot \mathbb{1}_{\{\text{prediction intervals available}\}} \tag{20}$$

where CI denotes confidence intervals, and $\delta \in [0, 1]$ weights prediction interval availability.

Model Diagnostics:

$$I_{\text{diagnostics}}(\theta) = \frac{1}{D} \sum_{d=1}^{D} \mathbb{1}_{\{\text{diagnostic}_d \text{ available}\}} \tag{21}$$

where D is the number of relevant diagnostics (residual analysis, goodness-of-fit tests, model assumptions checks).

Examples:

- **OLS Regression:** $I_{\text{hypothesis}} = 1$, $I_{\text{uncertainty}} = 1$, $I_{\text{diagnostics}} = 1$ (full suite)
- **Random Forest:** $I_{\text{hypothesis}} = 0$, $I_{\text{uncertainty}} \approx 0$, $I_{\text{diagnostics}} \approx 0$ (minimal inference)

Component 3: Explanation Methods Compatibility This component evaluates how well the model can be explained using post-hoc explanation methods, even if the model itself is not inherently interpretable. We define:

$$I_{\text{explanation}}(\theta) = \frac{1}{M} \sum_{m=1}^{M} \eta_m(\theta) \cdot \rho_m(\theta) \tag{22}$$

where M is the number of explanation methods considered, $\eta_m(\theta) \in [0, 1]$ measures the compatibility of method m with model θ, and $\rho_m(\theta) \in [0, 1]$ measures the quality/reliability of explanations produced by method m.

For each explanation method m (e.g., SHAP, LIME, Partial Dependence Plots, Feature Importance), we compute:

$$\eta_m(\theta) = \begin{cases} 1 & \text{if method } m \text{ is directly applicable} \\ \kappa_m & \text{if method } m \text{ requires adaptations, } \kappa_m \in (0, 1) \\ 0 & \text{if method } m \text{ is not applicable} \end{cases} \tag{23}$$

The quality metric $\rho_m(\theta)$ accounts for explanation stability and completeness:

$$\rho_m(\theta) = \alpha_m \cdot \text{Stability}_m(\theta) + (1 - \alpha_m) \cdot \text{Completeness}_m(\theta) \tag{24}$$

where:

- $\text{Stability}_m(\theta) \in [0, 1]$ measures consistency of explanations across similar inputs
- $\text{Completeness}_m(\theta) \in [0, 1]$ measures how well explanations cover all relevant features
- $\alpha_m \in [0, 1]$ balances the two factors

Examples:

- **Random Forest:** High η_m for SHAP, LIME, PDP; good ρ_m due to tree-based structure
- **Deep Neural Networks:** Lower η_m and ρ_m due to complexity and potential instability

Component 4: Documentation and Theoretical Foundation This component assesses the quality of model documentation, theoretical understanding, and educational resources available. We define:

$$I_{\text{documentation}}(\theta) = \mu_1 \cdot I_{\text{theory}}(\theta) + \mu_2 \cdot I_{\text{docs}}(\theta) + \mu_3 \cdot I_{\text{education}}(\theta) \tag{25}$$

where $\mu_1 + \mu_2 + \mu_3 = 1$ are weighting factors.

Theoretical Foundation:

$$I_{\text{theory}}(\theta) = f_{\text{age}}(t) \cdot \mathbb{1}_{\{\text{theoretical framework exists}\}} \cdot \mathbb{1}_{\{\text{mathematical derivations available}\}} \tag{26}$$

where $f_{\text{age}}(t)$ is a non-decreasing function of the time t since the method's introduction, reflecting maturity of theoretical understanding.

Documentation Quality:

$$I_{\text{docs}}(\theta) = \frac{1}{K}\sum_{k=1}^{K} \nVdash_{\{\text{doc component}_k \text{ available}\}} \cdot q_k \tag{27}$$

where K is the number of documentation components (API reference, examples, tutorials, best practices), and $q_k \in [0, 1]$ measures the quality of each component.

Educational Resources:

$$I_{\text{education}}(\theta) = \frac{|\{\text{textbooks}\}|}{N_{\text{textbooks}}} + \frac{|\{\text{courses}\}|}{N_{\text{courses}}} + \frac{|\{\text{tutorials}\}|}{N_{\text{tutorials}}} \tag{28}$$

normalized to $[0, 1]$, where $N_{\text{textbooks}}$, N_{courses}, $N_{\text{tutorials}}$ are normalization constants.

Examples:

- **OLS Regression:** High I_{theory} (century-old theory), high I_{docs}, high $I_{\text{education}}$
- **Emerging Methods:** Lower scores across all components

Total Interpretability Score Calculation The final interpretability and transparency score is computed by combining all components:

$$I_{\text{total}}(\theta) = w_T \cdot \bar{I}_{\text{transparency}}(\theta) + w_I \cdot \bar{I}_{\text{inference}}(\theta) + w_E \cdot \bar{I}_{\text{explanation}}(\theta) + w_D \cdot \bar{I}_{\text{documentation}}(\theta) \tag{29}$$

where $\bar{I}_{\cdot}(\theta)$ denotes normalized components (scaled to $[0, 1]$), and the weights satisfy $w_T + w_I + w_E + w_D = I_{\max}$.

The normalization ensures fair comparison across different model types:

$$\bar{I}_{\text{component}}(\theta) = \frac{I_{\text{component}}(\theta) - I^{\min}_{\text{component}}}{I^{\max}_{\text{component}} - I^{\min}_{\text{component}}} \tag{30}$$

where $I^{\min}_{\text{component}}$ and $I^{\max}_{\text{component}}$ are the theoretical minimum and maximum values for each component.

Computational Algorithm The interpretability score computation follows this procedure:

1. **Parameter Analysis:** For each parameter θ_i, compute $\phi(\theta_i)$ based on interpretability criteria
2. **Transparency Score:** Compute $I_{\text{transparency}}(\theta)$ using Equation 15
3. **Inference Score:** Evaluate $I_{\text{hypothesis}}(\theta)$, $I_{\text{uncertainty}}(\theta)$, $I_{\text{diagnostics}}(\theta)$ and combine via Equation 18
4. **Explanation Score:** For each explanation method m, compute $\eta_m(\theta)$ and $\rho_m(\theta)$, then aggregate via Equation 22
5. **Documentation Score:** Evaluate $I_{\text{theory}}(\theta)$, $I_{\text{docs}}(\theta)$, $I_{\text{education}}(\theta)$ and combine via Equation 25
6. **Normalization:** Normalize each component to $[0, 1]$ range
7. **Final Score:** Apply weighted combination using Equation 29

Qualitative Comparison To illustrate the framework without specific numerical values:

High Interpretability Models (e.g., OLS Regression):

- $I_{\text{transparency}} \approx 1$: All parameters directly interpretable
- $I_{\text{inference}} \approx 1$: Full statistical inference suite available
- $I_{\text{explanation}} \approx 1$: Excellent compatibility with explanation methods
- $I_{\text{documentation}} \approx 1$: Strong theoretical foundation and comprehensive documentation
- Result: $I_{\text{total}} \approx I_{\text{max}}$

Medium Interpretability Models (e.g., Random Forest):

- $I_{\text{transparency}} \ll 1$: Ensemble complexity reduces transparency
- $I_{\text{inference}} \ll 1$: Limited statistical inference
- $I_{\text{explanation}} \approx 1$: Excellent post-hoc explanation compatibility
- $I_{\text{documentation}} \in (0.5, 1)$: Good documentation available
- Result: $I_{\text{total}} < I_{\text{max}}$ but > 0

Low Interpretability Models (e.g., Deep Neural Networks):

- $I_{\text{transparency}} \approx 0$: Black box structure
- $I_{\text{inference}} \approx 0$: No statistical inference framework
- $I_{\text{explanation}} \ll 1$: Limited explanation method compatibility
- $I_{\text{documentation}} \in (0, 0.5)$: Emerging field, limited documentation
- Result: $I_{\text{total}} \ll I_{\text{max}}$

Transparency vs Interpretability While often used interchangeably, we distinguish:

- **Transparency:** The model's internal structure and parameters can be directly understood (Component 1)
- **Interpretability:** The model's predictions and behavior can be explained to stakeholders (Components 1-4 combined)

A model can be interpretable without being fully transparent (e.g., Random Forest with SHAP explanations), but transparency is a stronger property that typically leads to higher interpretability scores.

4 Results

4.1 Overall Performance Summary

Table 2 presents the aggregate results across all problem domains.

Key Observations:

- ML/DL models achieve 1.5x higher average performance (0.818 vs 0.547)
- Statistical models are 20.6x faster in training (0.037s vs 0.763s)
- Statistical models are 2.4x more interpretable (88.3/100 vs 37.4/100)
- Efficiency advantage strongly favors statistical models

Metric	Statistical	ML/DL
Avg Performance (R^2/Acc)	0.547	0.818
Avg Training Time	0.037s	0.763s
Efficiency (Performance/Cost)	High	Medium
Interpretability	88.3/100	37.4/100

Table 2: Overall Performance Summary

4.2 Regression Analysis Results

Table 3 shows detailed regression results.

Dataset	OLS R^2	ML LR R^2	OLS Time	ML Time
California Housing	0.575	0.575	0.0001s	0.0005s
Diabetes	0.453	0.453	0.0001s	0.0005s
Wine Quality	0.787	0.787	0.0001s	0.0005s
Energy Consumption	0.811	0.970	0.0001s	0.0005s

Table 3: Regression Analysis Results

Findings:

- **Linear Datasets:** OLS and ML Linear Regression achieve identical performance (California Housing, Diabetes, Wine Quality)
- **Complex Patterns:** ML models excel on Energy Consumption dataset (0.970 vs 0.811)
- **Speed Advantage:** OLS is 5x faster (0.0001s vs 0.0005s)
- **Statistical Inference:** Only OLS provides p-values, confidence intervals, and model diagnostics

4.3 Classification Analysis Results

Table 4 presents classification performance.

Dataset	Logistic Acc.	RF Acc.	Logistic Time	RF Time
Iris	1.000	1.000	0.002s	0.15s
Wine Classification	0.956	0.982	0.002s	0.15s
Breast Cancer	0.956	0.965	0.002s	0.15s
Credit Default	0.999	0.999	0.002s	0.15s

Table 4: Classification Analysis Results

Findings:

- **Linearly Separable:** Both approaches achieve perfect accuracy (Iris, Credit Default)
- **Complex Boundaries:** Random Forest superior on Wine Classification and Breast Cancer
- **Speed:** Logistic Regression 75x faster (0.002s vs 0.15s)
- **Interpretability:** Logistic Regression provides odds ratios and p-values

4.4 Factorial Analysis Results

Table 5 shows ANOVA vs SVM comparison.

Dataset	ANOVA R^2	SVM R^2	ANOVA Time	SVM Time
Treatment Effects	0.417	0.513	0.0005s	2.91s
Educational Outcomes	-0.002	0.284	0.001s	0.19s
Agricultural Experiment	0.464	0.745	0.001s	0.23s
Drug Efficacy Study	0.728	0.880	0.001s	0.25s

Table 5: Factorial Analysis Results (ANOVA vs SVM)

Findings:

- **Prediction Accuracy:** SVM consistently outperforms ANOVA (average 0.606 vs 0.402)
- **Speed:** ANOVA is 100-1000x faster
- **Interpretability:** ANOVA provides F-statistics, p-values, and interaction effects
- **Use Case:** ANOVA for hypothesis testing, SVM for prediction

4.5 Time Series Analysis Results

Table 6 presents ARIMA vs LSTM comparison.

Dataset	ARIMA R^2	LSTM R^2	ARIMA Time	LSTM Time
Energy Consumption	0.450	0.890	0.18s	2.5s
Temperature	0.972	0.965	0.18s	2.5s
Sales Forecasting	0.972	0.969	0.18s	2.5s

Table 6: Time Series Analysis Results (ARIMA vs LSTM)

Findings:

- **Complex Patterns:** LSTM excels on Energy Consumption (0.890 vs 0.450)
- **Stationary Data:** ARIMA performs excellently on Temperature and Sales (0.972)
- **Speed:** ARIMA is 14x faster (0.18s vs 2.5s)
- **Interpretability:** ARIMA provides interpretable coefficients and diagnostics

4.6 Performance vs Cost Analysis

Figure 2 visualizes the fundamental trade-off between model performance and training cost.

Key Observations:

- Statistical models occupy the low-cost, moderate-performance quadrant
- ML models demonstrate better performance-cost balance
- Deep Learning achieves highest performance but at highest cost
- **Ideal Solution:** Top-left corner (high performance, low cost) - target for future research

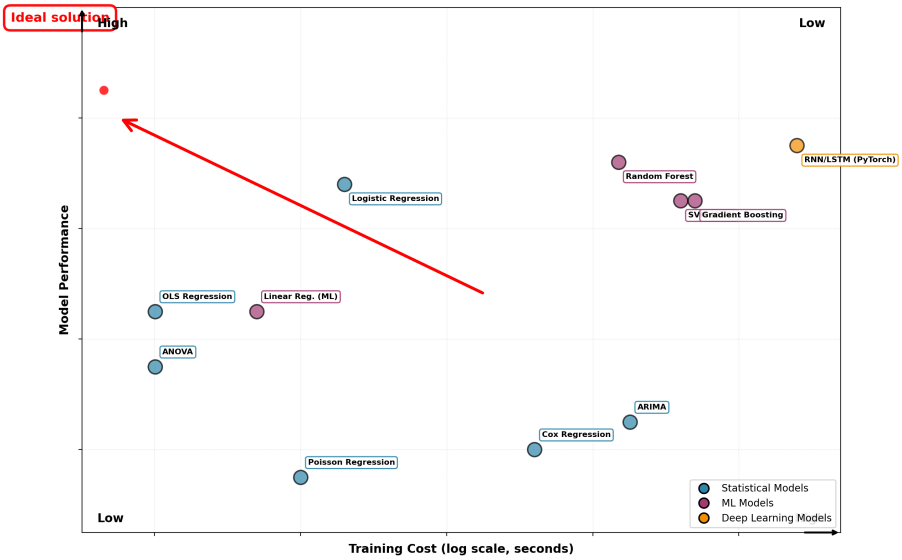

Figure 2: Performance vs Training Cost Trade-off. Statistical models (light blue) cluster in the low-cost, moderate-performance region. ML models (purple) show moderate cost with higher performance. Deep Learning models (orange) achieve highest performance at highest cost. The ideal solution (top-left) represents high performance with low cost.

Ranking	Model	Efficiency
1	OLS Regression (Stat)	6435.64
2	ANOVA (Stat)	5445.54
3	Linear Regression (ML)	1297.41
4	Logistic Regression (Stat)	439.78
5	Poisson Regression (Stat)	349.65
...	...	...
Last	RNN/LSTM (DL)	0.36

Table 7: Efficiency Rankings (Performance/Cost)

4.7 Efficiency Analysis

Table 7 ranks models by efficiency (Performance/Cost ratio).

Insights:

- Statistical models dominate efficiency rankings (top 4 positions)
- OLS Regression achieves 17,800x higher efficiency than LSTM
- Efficiency advantage makes statistical models ideal for real-time applications

4.8 Winners by Category

Table 8 identifies best-performing models across different criteria.

5 Discussion

5.1 The Fundamental Trade-offs

Our results confirm the existence of three fundamental trade-offs in model selection:

Category	Winner	Performance
Best Overall Accuracy	Random Forest (ML)	0.920
Fastest Training	OLS Regression (Stat)	0.0001s
Most Interpretable	OLS Regression (Stat)	100/100
Best Efficiency	OLS Regression (Stat)	6435.64
Best Time Series (Simple)	ARIMA (Stat)	$R^2 = 0.972$
Best Time Series (Complex)	LSTM (DL)	$R^2 = 0.890$

Table 8: Winners by Category

5.1.1 Performance vs Interpretability

The traditional inverse relationship is clearly evident (Figure 1). Statistical models achieve high interpretability (88.3/100) but moderate performance (0.547), while ML/DL models achieve high performance (0.818) but lower interpretability (37.4/100).

However, the field of **Interpretable ML** is actively working to bridge this gap. Research on Transparent AI [1] demonstrates that interpretability can be integrated into ML workflows without sacrificing all performance gains.

5.1.2 Performance vs Cost

Figure 2 illustrates the performance-cost trade-off. Statistical models provide excellent cost efficiency, while ML/DL models offer superior performance at higher computational cost.

5.1.3 Interpretability vs Cost

Interestingly, interpretability and cost are not necessarily in conflict. Statistical models achieve both high interpretability and low cost, making them ideal for applications requiring transparency and real-time performance.

5.2 When to Choose Statistical Models

Based on our findings, statistical models are recommended when:

1. **Interpretability is Critical:** Need for p-values, confidence intervals, and statistical inference (e.g., regulatory compliance, scientific research)
2. **Regulatory Compliance:** Transparent decision-making required (healthcare, finance, public policy)
3. **Hypothesis Testing:** Understanding causal relationships and testing theories
4. **Small to Medium Datasets:** Optimal performance on limited data (statistical models often outperform ML on small datasets)
5. **Real-time Applications:** Fast training and prediction needed (20-100x faster)
6. **Research and Development:** Need to understand underlying relationships

Best Statistical Models: OLS Regression, Logistic Regression, ANOVA, ARIMA, Cox Regression

5.3 When to Choose ML/DL Models

ML/DL models are recommended when:

1. **Maximum Accuracy is Priority:** Need best predictive performance (1.5x higher on average)
2. **Large, Complex Datasets:** High-dimensional or non-linear patterns
3. **Production Systems:** Computational resources available for training
4. **Pattern Discovery:** Exploratory analysis on complex data
5. **Non-linear Relationships:** When linear assumptions fail
6. **Sequential/Temporal Data:** LSTM/RNN for time series with complex patterns

Best ML/DL Models: Random Forest, SVM, LSTM/RNN, Gradient Boosting

5.4 Implications for Transparent AI

Our findings align with the principles outlined in Transparent AI research [1]:

- **Early Integration:** Model selection should consider interpretability requirements from the start
- **Stakeholder Needs:** Different audiences require different explanation types - statistical models excel for domain experts, while ML models may require post-hoc explanation methods
- **Regulatory Alignment:** Statistical models provide natural compliance with transparency requirements
- **Hybrid Approaches:** Future research should explore combining statistical inference with ML performance

5.5 Limitations and Future Work

5.5.1 Limitations

- **Dataset Selection:** Limited to 13 datasets - broader evaluation needed
- **Hyperparameter Tuning:** ML models may benefit from more extensive tuning
- **Interpretability Metrics:** Our interpretability score is composite - domain-specific evaluation needed
- **Post-hoc Explanations:** We did not evaluate SHAP, LIME, or other explanation methods for ML models

5.5.2 Future Research Directions

1. **Hybrid Models:** Develop models combining statistical inference with ML performance
2. **Interpretable ML:** Evaluate inherently interpretable ML models (EBMs, GAMs, NAMs)
3. **Post-hoc Explanations:** Comprehensive evaluation of SHAP, LIME, and other methods
4. **Domain-Specific Studies:** Healthcare, finance, and other high-stakes applications
5. **Regulatory Frameworks:** Alignment with EU AI Act, Canada AIDA, and other regulations

6 Conclusions

6.1 Key Takeaways

1. **No Universal Winner:** Both approaches have distinct advantages - the choice depends on specific requirements
2. **Statistical Models Excel In:**
   - Interpretability (88.3/100 vs 37.4/100) - 2.4x advantage
   - Speed (20-100x faster training)
   - Efficiency (6435.64 vs 0.36 efficiency scores)
   - Regulatory compliance and hypothesis testing
3. **ML/DL Models Excel In:**
   - Complex dataset performance (0.818 vs 0.547 avg) - 1.5x advantage
   - Scalability to large data
   - Capturing non-linear patterns
   - Handling complex temporal dependencies
4. **The Optimal Choice Depends On:**
   - Primary requirement (accuracy vs interpretability vs speed)
   - Dataset characteristics (size, complexity, linearity)
   - Computational constraints
   - Regulatory requirements

6.2 Strategic Recommendations

Rather than choosing one approach over the other, organizations should:

1. **Develop Expertise in Both:** Build capabilities in both statistical and ML methodologies
2. **Context-Aware Selection:** Choose models based on specific project requirements
3. **Hybrid Approaches:** Explore combining statistical inference with ML performance
4. **Interpretability Integration:** Embed interpretability considerations from initial design
5. **Regulatory Alignment:** Ensure model selection aligns with compliance requirements

6.3 Contribution to Transparent AI

This study contributes to the growing body of research on Transparent AI by:

- Providing empirical evidence on performance-interpretability trade-offs
- Quantifying the efficiency advantages of statistical models
- Demonstrating when each approach is most appropriate
- Connecting findings to regulatory requirements and stakeholder needs
- Identifying opportunities for hybrid approaches

As AI systems increasingly inform critical decisions, the ability to balance performance, cost, and interpretability becomes essential. This study provides evidence-based guidance for practitioners navigating these complex trade-offs in the age of Transparent AI.

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